

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2025

Course: B. Tech.

Branch: Civil Engineering

Semester: VIII

Subject Code & Name: BTCESS802C Remote Sensing Essentials

Max Marks: 60

Date:13/ 01/ 2026

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	Remote sensing is the science of obtaining information about objects on Earth by	CO1	1				
	<table border="1"> <tr> <td>A) Direct field measurement</td><td>B) Analysis of reflected or emitted electromagnetic energy</td><td>C) Ground survey only</td><td>D) Photographic mapping</td></tr> </table>	A) Direct field measurement	B) Analysis of reflected or emitted electromagnetic energy	C) Ground survey only	D) Photographic mapping		
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2	Wien's displacement law relates temperature of a body with	CO1	1				
	<table border="1"> <tr> <td>A) Total emitted energy</td><td>B) Peak wavelength of emission</td><td>C) Radiant intensity</td><td>D) Emissivity</td></tr> </table>	A) Total emitted energy	B) Peak wavelength of emission	C) Radiant intensity	D) Emissivity		
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3	Remote sensing images are represented digitally based on	CO1	1				
	<table border="1"> <tr> <td>A) Image projection</td><td>B) Pixel brightness values</td><td>C) Satellite altitude</td><td>D) Sensor orientation</td></tr> </table>	A) Image projection	B) Pixel brightness values	C) Satellite altitude	D) Sensor orientation		
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4	SPOT satellite is mainly known for	CO2	1				
	<table border="1"> <tr> <td>A) Microwave imaging</td><td>B) Higher spatial resolution</td><td>C) Thermal imaging only</td><td>D) Weather monitoring</td></tr> </table>	A) Microwave imaging	B) Higher spatial resolution	C) Thermal imaging only	D) Weather monitoring		
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		compared to LANDSAT				
5	Cartosat satellites are best suited for				CO2	1
	A) Oceanographic studies	B) Large-scale mapping and DEM generation	C) Atmospheric studies	D) Vegetation health analysis		
6	False colour composite uses which band for red colour?				CO2	1
	A) Blue band	B) Green band	C) Infrared band	D) Thermal band		
7	Ground control points are required mainly for				CO3	1
	A) Image classification	B) Georeferencing	C) Image compression	D) Radiometric correction		
8	High spatial resolution images are limited by				CO3	1
	A) Low data volume	B) High storage and processing requirement	C) Low accuracy	D) Poor coverage		
9	NDVI values range between				CO4	1
	A) 0 to 1	B) -1 to +1	C) 0 to 255	D) -255 to +255		
10	Lineament mapping is useful for				CO4	1
	A) Soil mapping	B) Groundwater potential zone identification	C) Vegetation analysis	D) Climate study		
11	GIS acts mainly as				CO5	1
	A) Sensor platform	B) Spatial data storage and analysis system	C) Image capture system	D) Photo viewer		
12	Ground truth data is required mainly for				CO5	1
	A) Image enhancement	B) Accuracy assessment	C) Compression	D) Radiometric calibration		
Q. 2	Solve the following.					12

A)	Explain the basic principle of remote sensing with suitable examples.	CO1	6
B)	Explain the effects of interaction of radiation on earth surface with respect to water and vegetation.	CO1	6
Q.3	Solve the following.		12
A)	Describe advantages of UAV in remote sensing applications.	CO2	6
B)	Explain various techniques of image acquisition in remote sensing.	CO2	6
Q. 4	Solve Any Two of the following.		12
A)	Write short notes on atmospheric correction.	CO3	6
B)	Explain the key stages in digital image processing for image compression.	CO3	6
C)	Write short notes on radiometric resolution.	CO3	6
Q.5	Solve Any Two of the following.		12
A)	Explain working of LiDAR technique and its applications in civil engineering.	CO4	6
B)	What are the major characteristic features of hyperspectral remote sensing and its applications?	CO4	6
C)	Write short notes on NDVI.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	What are the basic elements of image interpretation?	CO5	6
B)	What are the applications of RS and GIS in groundwater studies?	CO5	6
C)	Discuss challenges faced in interpretation of satellite images.	CO5	6
	*** End ***		